WORKED SOLUTION

STEP I 1995, Question 10

Mechanics • Hooke's Law

STEP I 1995, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s1-q10

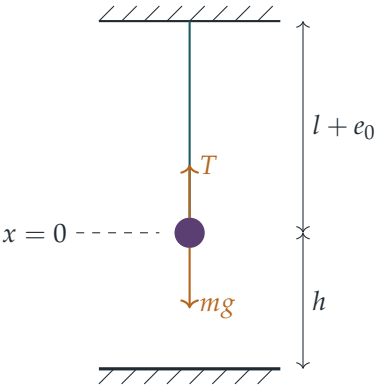
The key idea

Measured from the equilibrium position, the weight and the equilibrium part of the tension cancel exactly, so the only unbalanced force is a Hooke's-law restoring force proportional to the displacement x. Simple harmonic motion follows, and its energy equation is the speed formula. Everything after that is bookkeeping: the energy equation carries the ball between $x = 0$ and the surface, and Newton's experimental law reverses and scales the velocity at each bounce.

Solution

We model the ball as a particle and the string as light and elastic, obeying Hooke's law while it is stretched. We ignore air resistance, and take the bounce to be instantaneous. Throughout we write

$$\omega^2 = \frac{\lambda}{lm}.$$



The equilibrium length

Let the equilibrium extension be e_0, so the tension is $T_0 = \lambda e_0/l$. Resolving vertically for the ball in equilibrium, $T_0 = mg$, so

$$\frac{\lambda e_0}{l} = mg \quad \Longrightarrow \quad e_0 = \frac{mgl}{\lambda},$$

and the total length is $l + e_0 = l(1 + mg/\lambda)$, as required.

The speed formula

Let x be the distance of the ball below the equilibrium position, with downwards positive. For $x \geq 0$ the extension is $e_0 + x > 0$, so the string is taut with tension $\lambda(e_0 + x)/l$. Newton's second law downwards gives

$$m\ddot{x} = mg - \frac{\lambda(e_0 + x)}{l} = -\frac{\lambda x}{l},$$

because $\lambda e_0/l = mg$. So $\ddot{x} = -\omega^2 x$: simple harmonic motion about $x = 0$.

Writing $\ddot{x} = v\,\dfrac{\mathrm{d}v}{\mathrm{d}x}$ and integrating,

$$\int v\,\mathrm{d}v = -\omega^2 \int x\,\mathrm{d}x \quad \Longrightarrow \quad v^2 = -\omega^2 x^2 + C.$$

At $x = 0$ the speed is u_0, so $C = u_0^2$ and

$$v^2 + \frac{\lambda}{lm}x^2 = u_0^2,$$

as required. (Equivalently, this is conservation of energy: the gravitational and elastic energies combine into $\frac{1}{2} \cdot \frac{\lambda}{l}x^2$ plus a constant.)

The first bounce

The ball reaches the surface only if $v^2 > 0$ at $x = h$, that is, if $u_0^2 - \omega^2 h^2 > 0$. This is exactly the given condition $\lambda h^2 < lmu_0^2$. So the ball strikes the surface with speed w_0, where

$$w_0^2 = u_0^2 - \omega^2 h^2.$$

By Newton's experimental law, it leaves the surface moving upwards with speed ew_0.

On the way up from $x = h$ to $x = 0$ the string is still stretched, so the same equation of motion holds and the energy integral is $v^2 + \omega^2 x^2 = \text{constant}$. Only the constant is new. At $x = h$ we have $v^2 = e^2 w_0^2$, so the constant is $e^2 w_0^2 + \omega^2 h^2$, and at $x = 0$

$$\begin{aligned} u_1^2 &= e^2 w_0^2 + \omega^2 h^2 \\ &= e^2\left(u_0^2 - \omega^2 h^2\right) + \omega^2 h^2 \\ &= e^2 u_0^2 + \frac{\lambda}{lm}h^2\left(1 - e^2\right), \end{aligned}$$

as required. Here u_1 is directed upwards.

The second bounce

Ask yourself After passing $x = 0$ the ball rises above the equilibrium position, and the string may go slack. Does the ball still come back to $x = 0$ with speed u_1?

Yes. Above $x = 0$ the ball moves under its weight and the string tension, or under gravity alone if the string goes slack. Both forces are conservative: when the string is slack its elastic energy is simply zero. So the total mechanical energy is conserved for the whole of the flight above $x = 0$. When the ball comes back down to $x = 0$ it has the same potential energy as when it left, and therefore the same speed u_1, now directed downwards.

So the second descent is exactly like the first, with u_1 in place of u_0. We still need the ball to reach the surface, and it does, because

$$u_1^2 - \omega^2 h^2 = e^2 \left(u_0^2 - \omega^2 h^2 \right) > 0.$$

Repeating the first-bounce argument with u_0 replaced by u_1,

$$\begin{aligned} u_2^2 &= e^2 u_1^2 + \omega^2 h^2 \left(1 - e^2\right) \\ &= e^2 \left[e^2 u_0^2 + \omega^2 h^2 (1 - e^2) \right] + \omega^2 h^2 (1 - e^2) \\ &= e^4 u_0^2 + \omega^2 h^2 (1 - e^2)(1 + e^2) \\ &= e^4 u_0^2 + \frac{\lambda}{lm} h^2 \left(1 - e^4\right), \end{aligned}$$

which completes the question.

Checks

- If $e = 1$ there is no energy loss, and both formulae give $u_1 = u_2 = u_0$, as they should.
- If $e = 0$ the ball leaves the surface at rest, and then $u_1^2 = \omega^2 h^2$: the ball rises from rest at amplitude h and passes the centre at speed ωh, as simple harmonic motion predicts.
- A neat way to see the pattern: $u_n^2 - \omega^2 h^2 = e^{2n} \left(u_0^2 - \omega^2 h^2 \right)$, because each bounce multiplies the squared impact speed by e^2.
- We stepped the motion numerically (Runge–Kutta, with the string allowed to go slack). For example, with $m = 1$, $l = 1$, $\lambda = 20$, $g = 9.8$, $u_0 = 6$, $h = 0.3$ and $e = 0.9$ the simulation gives $u_1 = 5.4316$ and $u_2 = 4.9233$, matching the formulae. In this example the string does go slack between bounces.

Common slips

- Measuring x from the natural length rather than from equilibrium. Then the weight doesn't cancel, and the formula comes out with extra linear terms.
- Applying the restitution factor e to the speed at $x = 0$ instead of to the impact speed at $x = h$.
- Assuming the string stays taut above equilibrium. It needn't, but energy conservation makes this irrelevant to the speed at $x = 0$.

Beyond STEP

The cancellation of gravity about the equilibrium point is a first example of a general principle: linearise about an equilibrium, and small oscillations are simple harmonic with ω^2 equal to the "stiffness" divided by the mass. The bouncing ball is an example of a piecewise-smooth or *impact* oscillator. These systems can behave remarkably richly (period doubling and even chaos) when the surface itself is moving. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats oscillators and energy methods at first-year level.

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